# Spectral Optimization Across Vision Architecture Families: A Controlled Empirical Study

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## Abstract

Spectral optimizers like Muon achieve striking gains over AdamW in vision tasks, but what drives their advantage and when does it break down? We present a controlled empirical study comparing Muon across three architectures (ResNet-50, ConvNeXt-T, DeiT-S) and three datasets, with SOAP and Shampoo compared on CIFAR-100 (and Tiny-ImageNet in the appendix), that reveals that the spectral advantage holds across all tested architectures, datasets, and training lengths; for Muon, Newton–Schulz orthogonalization is the primary contributor. Removing it collapses performance by 17 pp, erasing the entire spectral advantage and confirming that spectral preconditioning, not momentum or learning-rate scheduling, is the primary component. This orthogonalization reshapes weight spectra toward uniform singular-value distributions, and layer-wise spectral divergence between optimizers correlates with per-layer accuracy impact on CNNs, providing a mechanistic rather than purely empirical explanation. A normalization-swap experiment yields the most striking result: replacing batch normalization with layer normalization causes AdamW to collapse (67.85% $\rightarrow$ 50.51%) while Muon is unaffected (76.88% $\rightarrow$ 78.52%), demonstrating that spectral optimizers exhibit robustness to normalization changes compared to first-order methods. Muon converges 2–15$\times$ faster than AdamW in wall-clock time across CIFAR-100 and Tiny-ImageNet. These findings establish that spectral structure control is the primary driver of optimizer robustness across normalization regimes in modern vision architectures, particularly on convolutional architectures[1].

## 1 Introduction

Modern vision architectures have diverged rapidly in structural design: vision transformers use layer normalization and global self-attention Touvron et al. [2021], hybrid architectures combine depthwise separable convolutions with layer normalization Liu et al. [2022], while classical CNNs retain batch normalization He et al. [2016]. Yet optimizer selection has not adapted accordingly; practitioners still default to AdamW or SGD regardless of these structural differences, treating optimizer choice as secondary to architecture and data.

This neglect is consequential. SGD collapses to $11.51\%$ accuracy on layer-normalized ConvNeXt-T while achieving $69.50\%$ on batch-normalized ResNet-50, suggesting normalization type as a critical factor, a hypothesis we later confirm causally (Section 5.3). The same task remains accessible to other optimizers, revealing that the optimizer-architecture interaction is not a minor hyperparameter issue but a fundamental compatibility constraint that current practice ignores.

[1]Code is available at: https://anonymous.4open.science/r/spectral_optim_vision_geometry-DC6A

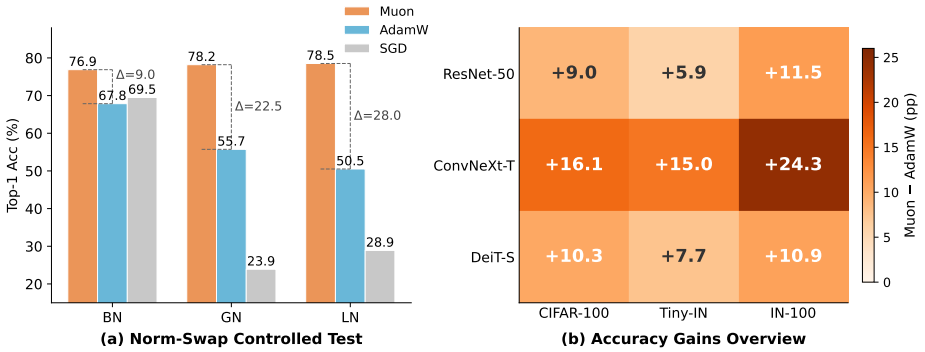



Figure 1: Overview of spectral optimization advantages in vision tasks. (a) Controlled norm-swap experiment: Muon's advantage persists across BatchNorm, GroupNorm, and LayerNorm ($\Delta < 2$ pp), while AdamW degrades sharply without BatchNorm. (b) Accuracy gains (Muon − AdamW, pp) across architectures and datasets.

Why do some optimizers transfer across architectures while others catastrophically fail? Existing first-order methods (AdamW with diagonal curvature estimates, SGD with raw gradient direction) do not explicitly account for how architectural design reshapes the spectral properties of weight gradients. Prior optimizer comparisons typically evaluate single architectures Schmidt et al. [2021] or lack controlled protocol isolation Choi et al. [2020], making it impossible to disentangle optimizer capability from recipe engineering.

We hypothesize that optimizers exploiting *weight-matrix spectral structure*, through Newton–Schulz orthogonalization or Kronecker-factored preconditioning Bernstein and Newhouse [2024], should exhibit architecture-robust performance precisely because they adapt to the spectral geometry regardless of how normalization and layer composition reshape the spectral geometry. We test this through a controlled study of three spectral optimizers (Muon Jordan et al. [2024][2], SOAP Vyas et al. [2025], Shampoo Gupta et al. [2018]) against AdamW Loshchilov and Hutter [2019] and SGD, spanning three architecture families, three datasets, and two training horizons, with independently tuned learning rates per optimizer-architecture pair (Section 4.2).

Our experiments reveal three main findings. First, spectral optimization is a *family-level* property: all three methods outperform AdamW by $+9$ to $+16$ percentage points on CIFAR-100 where all were evaluated, with Muon maintaining its advantage ($+5.9$ to $+24.3$ pp) across all three datasets. Second, the advantage is architecture-dependent (largest on layer-normalized networks, smallest on batch-normalized ones), suggesting normalization type mediates how spectral preconditioning benefits optimization. Third, spectral methods reach near-ceiling accuracy within 20 epochs, surpassing AdamW's 50-epoch performance at roughly half the wall-clock cost.

**Contributions.**

- The first controlled empirical validation of the spectral optimizer family hypothesis Bernstein and Newhouse [2024]: all three methods (Muon, SOAP, Shampoo) confer a consistent advantage across three vision architectures and three datasets, establishing optimizer-architecture coupling as a structural design concern.
- Causal evidence that the spectral advantage is mediated by normalization type: replacing BatchNorm with LayerNorm on ResNet-50 collapses AdamW by $\sim$17 pp while Muon is unaffected, and layer-wise stable-rank-divergence correlations reverse sign between CNNs ($\rho$=$+0.45$) and transformers ($\rho$=$-0.40$).
- Practical guidance: spectral methods are architecture-robust candidates across the training conditions tested, achieving superior accuracy at substantially lower training cost.

[2]At the time of writing, the Muon reference is a blog post and has not been peer-reviewed.

## 2 Related Work

**Optimizer-architecture coupling.** The observation that optimizer performance depends on architecture is well-established but poorly addressed. SGD dominated the convolutional era He et al. [2016] but fails on modern layer-normalized architectures Liu et al. [2022], Touvron et al. [2021], driving a wholesale shift to AdamW Loshchilov and Hutter [2019]. Large-scale benchmarks confirm that no single optimizer dominates across architectures Choi et al. [2020], Schmidt et al. [2021], and Li et al. Li et al. [2024] formalize this as *backbone–optimizer coupling bias*, showing that optimizer rankings shift substantially with backbone choice. The underlying mechanism involves distinct optimization landscapes: batch normalization smooths loss surfaces in ways that layer normalization does not Santurkar et al. [2018], and Vision Transformers converge to sharper minima than CNNs Chen et al. [2022]. However, these studies diagnose the problem without offering a solution: they characterize *which* optimizer works where, but not *why* certain methods transfer and others fail. Our work addresses this gap by evaluating spectral optimizers, absent from prior benchmarks, and by providing layer-wise analysis that reveals where in the network coupling originates.

**Spectral and preconditioned optimization.** A parallel research thread has developed optimizers that exploit gradient matrix structure beyond diagonal curvature estimates. K-FAC Martens and Grosse [2015] pioneered Kronecker-factored natural gradient; Shampoo Gupta et al. [2018], Anil et al. [2021] extended this to full-matrix preconditioning with practical scalability; SOAP Vyas et al. [2025] introduced data-driven eigenbasis adaptation; and Muon Jordan et al. [2024] takes a distinct approach via Newton–Schulz orthogonalization, performing steepest descent under the spectral norm (Appendix F). Bernstein & Newhouse Bernstein and Newhouse [2024] unify these methods theoretically, proving that Adam, Shampoo, and Muon each correspond to steepest descent under different matrix norms, while Pethick et al. Pethick et al. [2025] formalize this further through norm-constrained optimization oracles. This unification motivates treating them as a *spectral optimizer family*, yet no prior work evaluates whether the family shares empirical properties (such as architecture robustness) beyond theoretical structure. Our study provides this evaluation across diverse vision architectures.

**Complementary optimizer directions.** Other recent optimizers target computational cost or tuning difficulty (Lion Chen et al. [2023] via program search, Sophia Liu et al. [2024] via diagonal Hessian clipping, Prodigy Mishchenko and Defazio [2024] via learning-rate-free estimation, GaLore Zhao et al. [2024] via low-rank projection) rather than architecture robustness. Our focus is complementary: which structural property determines whether an optimizer transfers across architectural designs.

## 3 Spectral Analysis Framework

### 3.1 Spectral Optimization via Orthogonalized Updates

Muon Jordan et al. [2024], Liu et al. [2025] optimizes 2D weight matrices by orthogonalizing the gradient before applying the update, effectively performing a Newton-like step in the spectral domain. Given a gradient matrix $G \in \mathbb{R}^{m\times n}$ for a weight matrix $W$, the goal is to compute an approximation to $(GG^\top)^{-1/2}G$, which projects the gradient onto the Stiefel manifold of orthonormal row matrices. Rather than forming this matrix inverse explicitly, Muon uses Newton–Schulz iteration to approximate it efficiently. Five iterations of a quintic Newton–Schulz polynomial orthogonalize $G$ (starting from $X_0 = G/\|G\|_F$; iteration formula, coefficients, and convergence analysis in Appendix F). The orthogonalized gradient serves as the update direction, combined with Nesterov momentum ($\beta = 0.95$).

This procedure captures inter-parameter curvature structure that first-order methods miss. SGD updates along the raw gradient direction; AdamW applies element-wise scaling via running second-moment estimates, which amounts to a diagonal approximation to the curvature. Muon's orthogonalization operates on the full gradient matrix, providing a non-diagonal second-order approximation related to K-FAC Martens and Grosse [2015], but without explicitly maintaining Kronecker-factored estimates. As described in Section 4, Muon is applied exclusively to 2D weight matrices (convolution kernels reshaped to 2D and linear layers), while remaining parameters are handled by a separate AdamW instance (Appendix Table 5 details the parameter routing).

### 3.2 Layer-wise Spectral Analysis Framework

Global accuracy differences between optimizers do not reveal *where* in the network one optimizer outperforms another. We introduce two complementary analysis tools that localize optimizer effects to individual layers.

**Weight Swap Protocol.** Given a Muon-trained model with weights $\theta^{\text{Muon}} = \{W_l^M\}$ and an AdamW-trained model with weights $\theta^{\text{AdamW}} = \{W_l^A\}$, we construct a hybrid model $\theta^{\text{Muon}\setminus l \leftarrow \text{AdamW}_l}$ by replacing the weights of a single layer $l$ in the Muon model with the corresponding AdamW weights, keeping all other layers at their Muon-trained values. We define the per-layer Muon benefit:

$$\Delta a_l = \text{acc}(\theta^{\text{Muon}}) - \text{acc}(\theta^{\text{Muon}\setminus l \leftarrow \text{AdamW}_l}), \tag{1}$$

where $\text{acc}(\cdot)$ denotes validation accuracy. $\Delta a_l > 0$ indicates that Muon's training at layer $l$ outperforms AdamW; $\Delta a_l < 0$ indicates the reverse. The beneficial layer ratio, the fraction of layers where $\Delta a_l > 0$, aggregates this across all eligible layers.

**Spectral Correlation Analysis.** To connect the weight swap results with spectral structure, we measure how each optimizer shapes the *stable rank* of each layer's weight matrix:

$$\text{sr}(W_l) = \frac{\|W_l\|_F^2}{\sigma_1(W_l)^2}, \tag{2}$$

where $\sigma_1(W_l)$ is the largest singular value. Stable rank captures the effective dimensionality of the weight matrix: higher values indicate a more uniform singular value distribution, meaning the matrix spreads its representational capacity across more directions. We choose stable rank over alternatives (effective rank, condition number, spectral gap) because it is bounded, well-defined for rank-deficient matrices, and permits direct cross-architecture comparison without architecture-specific normalization. For convolutional layers, we reshape the 4D kernel tensor to a 2D matrix before computing $\text{sr}(W_l)$. We compute the per-layer stable rank divergence $\Delta\text{sr}_l = \text{sr}(W_l^{\text{Muon}}) - \text{sr}(W_l^{\text{AdamW}})$: positive values indicate Muon spreads spectral energy more uniformly (higher effective rank), negative values indicate Muon concentrates energy into fewer directions. We evaluate the Spearman rank correlation between $\Delta\text{sr}_l$ and $\Delta a_l$ (Eq. 1); results are reported in Section 5.3.

### 3.3 Evaluation Metrics and Statistical Protocol

We report best validation accuracy (early-stopping checkpoint) throughout, in percentage points (pp), to decouple optimizer comparison from convergence speed. Key experiments use 3–4 seeds (from $\{42, 0, 1, 2\}$; Appendix Table 11), reporting mean $\pm$ std (ddof $= 1$). Independent 50-epoch runs with cosine annealing ($T_{\max} = 50$) and the same per-pair learning rates test gap persistence without schedule-length confounding.

## 4 Experimental Setup

### 4.1 Datasets and Architectures

We evaluate on three benchmarks: CIFAR-100 Krizhevsky and Hinton [2009] (60K images, 100 classes, $32\times32$ resized to $224\times224$), Tiny-ImageNet Le and Yang [2015] (100K images, 200 classes, $64\times64$), and ImageNet-100 Deng et al. [2009] (a 100-class subset at native $224\times224$). Training augmentation uses `Resize(256)`, `RandomCrop(224)`, and random horizontal flip.

We select three architectures with comparable parameter counts (22–29M) spanning the major design paradigms: ResNet-50 He et al. [2016] (batch normalization, $\sim$25.6M), DeiT-S Touvron et al. [2021] (layer normalization, multi-head attention, $\sim$22M), and ConvNeXt-T Liu et al. [2022] (layer normalization, depthwise separable convolutions, $\sim$28.6M). All models are trained from scratch with He initialization He et al. [2015] using `timm` Wightman [2019] without pretrained weights.

### 4.2 Optimization Protocol

We compare five optimizers: Muon Jordan et al. [2024] (spectral-norm orthogonalization via Newton–Schulz iteration, Nesterov $\beta$=0.95, applied to 2D weight matrices with a companion AdamW for

remaining parameters), SOAP Vyas et al. [2025] (Kronecker-factored eigenbasis preconditioning), Shampoo Gupta et al. [2018] (full-matrix preconditioning), AdamW Loshchilov and Hutter [2019] ($\beta_1$=0.9, $\beta_2$=0.999, wd 0.05), and SGD Sutskever et al. [2013] (Nesterov momentum 0.9, wd $10^{-4}$). All configurations use batch size 128, cosine annealing, and 20 epochs for the main comparison with 50-epoch extensions for convergence analysis.

To ensure fair comparison, each optimizer-architecture-dataset combination receives an independent 5-epoch learning rate sweep over 3–7 log-spaced rates Schmidt et al. [2021]. Of 27 optimizer-architecture-dataset configurations, 18 confirmed a clear peak (accuracy declining at adjacent rates); 7 remained monotonic within the swept range (e.g., Muon on ConvNeXt-T at all three datasets, Shampoo on ResNet-50); and 2 used literature defaults without a formal sweep (Muon on ResNet-50 CIFAR-100, lr=0.02; SGD on ResNet-50, lr=0.1). For monotonic cases, we selected the highest-accuracy learning rate and validated the choice via 20-epoch saturation experiments; further increases either diverged or yielded no meaningful accuracy gain (Appendix Tables 7–9). The monotonic-improvement rate of 7/9 configurations (all involving spectral optimizers) suggests that 20-epoch gaps are, if anything, a conservative proxy for the spectral advantage, since the optimal learning rate may lie beyond the swept range. Selected learning rates and complete optimizer configurations are reported in Appendix A.

### 4.3 Evaluation and Reproducibility

We report best validation accuracy across all epochs (i.e., the early-stopping checkpoint) and quantify optimizer gaps in percentage points (pp). Key experiments are replicated with 3 seeds, reporting mean $\pm$ std. Independent 50-epoch runs verify whether gaps persist at longer horizons. All results are obtained on a single NVIDIA RTX 5090 GPU with PyTorch and `timm`; code and training logs will be released upon publication.

## 5 Results

We organize results around five questions: (1) whether the spectral advantage is family-level and how it depends on architecture (§5.1), (2) whether it persists with longer training (§5.2), (3) what layer-wise mechanisms underlie it (§5.3), (4) whether it generalizes to modern training recipes (§5.4), and (5) whether the advantage is attributable to spectral preconditioning itself (§5.5).

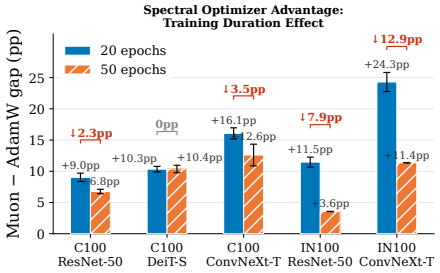
### 5.1 Spectral Advantage: Magnitude and Architecture Dependence

If spectral optimizers succeed by exploiting weight-matrix structure, their advantage should be universal across architectures but variable in magnitude. Table 1 confirms this prediction: Muon outperforms AdamW across all nine architecture-dataset pairs, with gaps ranging from $+5.9$ pp (ResNet-50/Tiny-ImageNet) to $+24.3$ pp (ConvNeXt-T/ImageNet-100), as visualized in Figure 1(b). Multi-seed variance is low (0.05–2.17 pp; Appendix Table 11), confirming that observed gaps substantially exceed run-to-run noise. All nine Muon-AdamW gaps are statistically significant after Holm–Bonferroni correction (Welch's $t$-test, $p < 0.02$, Cohen's $d > 10$ in every pair), confirming that the advantage is not attributable to seed variance. The 20-epoch budget serves as a controlled stress test of optimizer-intrinsic convergence capacity, independent of augmentation and scheduling. Our main comparison uses a fixed minimal recipe to isolate optimizer effects; §5.4 and Appendix show the advantage persists under enhanced recipes and independent per-optimizer LR sweeps, though fully architecture-optimized protocols may narrow the gap.

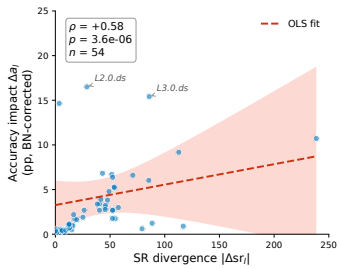
**Spectral Family Comparison on CIFAR-100.** A critical question for interpretation is whether the advantage is specific to Muon's Newton–Schulz mechanism or shared across spectral methods. We evaluate SOAP Vyas et al. [2025] and Shampoo Gupta et al. [2018] on CIFAR-100 under the same protocol (Table 1; full comparison in Appendix Table 19). SOAP matches Muon within noise on ResNet-50 and ConvNeXt-T and outperforms it on DeiT-S ($+3.10$ pp, $p < 0.001$). Within the spectral optimizer class, Shampoo leads on ResNet-50 ($+1.37$ pp over Muon), SOAP on DeiT-S ($+3.10$ pp, $p<0.001$), and Muon on ConvNeXt-T ($+4.05$ pp), with each architecture favoring a different spectral variant. All spectral optimizers outperform AdamW by $+9$–$+16$ pp on CIFAR-100, with intra-family differences (1.4–4.1 pp) approximately 4–8$\times$ smaller than the spectral-AdamW gap, suggesting the benefit as a *family-level* trend on CIFAR-100, with intra-class variation (Appendix A).

Table 1: Best validation accuracy (%) at 20 epochs. Multi-seed results (± std) where available; †single-seed (seed = 42). Bold = best per column. Gap = best spectral − AdamW in pp. ‡Muon−AdamW only (SOAP/Shampoo evaluated on CIFAR-100 only). Significance: Welch's $t$-test with Holm–Bonferroni correction across 2–4 seeds. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. For Tiny-ImageNet and ImageNet-100, where only Muon is evaluated among spectral methods, the gap is Muon−AdamW.

| Dataset | Optimizer | ResNet-50 | DeiT-S | ConvNeXt-T |
|---|---|---|---|---|
| CIFAR-100 | Shampoo | **78.25**±0.18 | 63.53±0.05 | 70.57±0.29 |
| | SOAP | 77.28±0.20 | **65.45**±0.18 | 74.40±0.43 |
| | Muon | 76.88±0.27$^{***}$ | 62.67±0.41$^{***}$ | **74.62**±0.15$^{**}$ |
| | AdamW | 67.85±0.17 | 52.33±0.34 | 58.54±2.17 |
| | SGD | 69.50±0.39 | 43.98±0.48 | 11.51† |
| | *Gap* | **+10.40** | **+13.12** | **+16.08** |
| Tiny-ImageNet | Muon | **62.66**±0.32$^{***}$ | **48.02**±0.21$^{***}$ | **58.92**±0.19$^{***}$ |
| | AdamW | 56.76±0.45 | 40.33±0.42 | 43.92±0.29 |
| | SGD | 60.04† | 34.22† | 9.49† |
| | *Gap*‡ | **+5.90** | **+7.69** | **+15.00** |
| ImageNet-100 | Muon | **61.97**±0.25$^{**}$ | **37.04**±0.38$^{***}$ | **57.83**±0.21$^{**}$ |
| | AdamW | 50.50±0.76 | 26.12±0.10 | 33.53±1.51 |
| | SGD | 50.06† | 20.38† | 6.77† |
| | *Gap*‡ | **+11.47** | **+10.92** | **+24.30** |




(a) Gap evolution from 20 to 50 epochs




(b) Layer-wise spectral divergence vs. accuracy impact

Figure 2: Temporal persistence and mechanistic basis of the spectral advantage. (a) Muon-AdamW gap narrows on CNNs but persists on DeiT-S, revealing convergence-speed vs. solution-quality mechanisms. (b) On ResNet-50, stable-rank divergence correlates with weight-swap accuracy impact ($\rho = +0.58$, $p = 3.6 \times 10^{-6}$), indicating broad spectral reshaping across layers.

**Architecture dependence.** Figure 1(b) reveals a consistent ordering: ConvNeXt-T exhibits the largest gap at every dataset (+16.08, +15.00, +24.30 pp), 1.2–2.5× larger than the other architectures, while ResNet-50 shows the smallest. ConvNeXt-T consistently shows the largest gap while ResNet-50 shows the smallest, a pattern consistent across all three spectral methods (Appendix Table 19). We discuss the mechanistic basis for this ordering in Section 6.

### 5.2 Persistence of the Spectral Advantage: 20 vs. 50 Epochs

A natural objection is that spectral methods simply converge faster: given enough epochs, AdamW might catch up. We extend training to 50 epochs on CIFAR-100 and ImageNet-100 (Table 2; Figure 2; epoch-level details in Appendix Table 14).

The answer is architecture-dependent. On ResNet-50, AdamW gains +3.16 pp while Muon gains +0.89 pp, narrowing the gap from +9.03 to +6.76 pp, a convergence-speed difference. On DeiT-S, both gains are negligible ($< 0.25$ pp) and the gap is unchanged (+10.37 vs. +10.34 pp), indicating a genuine solution-quality advantage. On ConvNeXt-T, the gap narrows from +16.08 to +12.60 pp as AdamW improves while Muon stays flat. The pattern reveals two mechanisms: *convergence speed* on CNNs (gap narrows but remains substantial) and *solution quality* on transformers (gap unchanged, both optimizers plateau early at different solutions). Spectral methods incur 5–22% per-epoch overhead but achieve equivalent accuracy in ≤20 epochs vs. ≥50 for AdamW, yielding net ≥2× wall-clock savings (Appendix Table 17).

Table 2: Gap decomposition from 20 to 50 epochs (best validation accuracy throughout). $\Delta$Gap = 50ep gap$-$20ep gap. Superscripts denote seed count: $^{(k)}$ = $k$-seed mean; $^{\dagger}$ = single seed (seed = 42). IN100 AdamW seed = 42 used lr=$3\times10^{-4}$ (vs. $3\times10^{-3}$ for other seeds); a verification run confirmed identical accuracy at both rates. All gaps are Muon $-$ AdamW.

| Setting | 20ep gap | 50ep gap | $\Delta$Gap | Muon $\Delta$ | AdamW $\Delta$ |
|---|---|---|---|---|---|
| C100 ResNet-50 | $+9.03^{(4)}$ | $+6.76^{(3)}$ | $-2.27$ | +0.89 | +3.16 |
| C100 DeiT-S | $+10.34^{(4)}$ | $+10.37^{(3)}$ | +0.03 | +0.24 | +0.21 |
| C100 ConvNeXt-T | $+16.08^{(4)}$ | $+12.60^{(3)}$ | $-3.48$ | $-0.03$ | +3.45 |
| IN100 ResNet-50 | $+11.47^{(4)}$ | $+3.56^{\dagger}$ | $-7.91$ | +0.27 | +8.18 |
| IN100 ConvNeXt-T | $+24.30^{(4)}$ | $\approx+11.37^{\dagger}$ | $\approx-12.93$ | +0.52 | +13.45 |

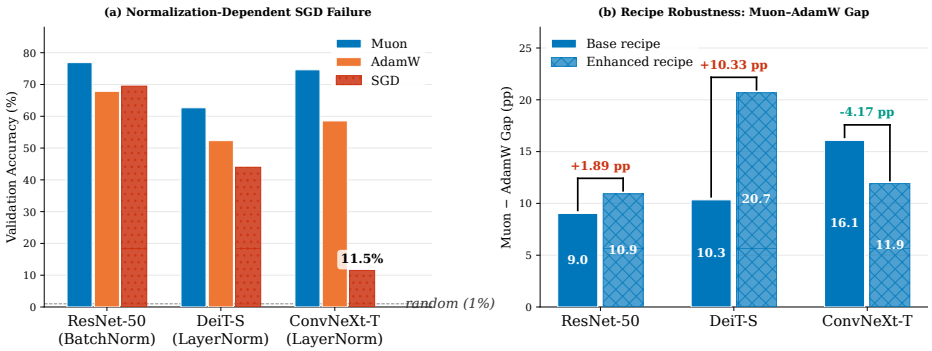



Figure 3: (a) Normalization-swap experiment on ResNet-50: replacing BatchNorm with LayerNorm collapses AdamW by $\sim$17 pp and SGD degrades catastrophically (28.87%, a $-40.6$ pp drop from BN), while Muon is preserved (+1.64 pp). (b) Training recipe robustness: enhanced augmentation produces differential optimizer response across architectures, where Muon gains while AdamW is flat or declines.

### 5.3 Mechanistic Analysis: Architecture-Dependent Correlation and Normalization Dependence

The preceding sections establish *that* spectral methods outperform across architectures; we now investigate *where* in the network this advantage arises and *why* it depends on normalization type.

**Layer-wise weight swap reveals architecture-dependent correlation.** Weight swap experiments (Section 3) show that Muon's advantage is near-universal on ResNet-50 (94.3% of layers beneficial) but selective on DeiT-S (51.7%). The Spearman correlation between per-layer stable rank divergence $\Delta\mathrm{sr}_l$ and per-layer Muon benefit $\Delta a_l$ is positive on ResNet-50 ($\rho = +0.58$, $p = 3.6 \times 10^{-6}$, after BN recalibration), as visualized in Figure 2. On DeiT-S, the layer-wise benefit pattern shows no significant stable-rank correlation ($\rho \approx 0$, n.s.), contrasting with ResNet-50. We report stable-rank divergence as our primary metric; the architecture-dependent correlation pattern is discussed further in Appendix J.

**Architecture-dependent mechanisms.** The correlation pattern reveals architecture-dependent mechanisms (Figure 2). On ResNet-50, the positive correlation ($\rho = +0.58$) means layers where Muon reshapes spectral structure most are where it helps most: *broad spectral reshaping* across the network, with 94.3% of layers benefiting. Muon-trained ConvNeXt-T models exhibit $7.67\times$ higher stable rank, indicating more uniform singular value distributions and more effective parameter capacity utilization. On DeiT-S, no significant stable-rank correlation is observed ($\rho \approx 0$, n.s.); the transformer mechanism remains to be characterized.

**SGD failure as extreme normalization dependence.** Figure 3 illustrates the most extreme manifestation: SGD collapses to near-random accuracy on ConvNeXt-T (11.51%; Appendix E) and degrades substantially on DeiT-S (43.98%), while remaining competitive on batch-normalized ResNet-50 (69.50%). Independent verification confirms this is not a configuration error: lr sweep across 5 rates and weight-decay sanity check ($\Delta = -0.09$ pp) both confirm SGD's genuine failure mode on

Table 3: Dose-response: accuracy (%) under different normalization types on ResNet-50, CIFAR-100.

| Optimizer | R50-BN | R50-GN(32) | R50-LN | BN→GN32 Δ |
|---|---|---|---|---|
| Muon | 76.88 | 78.20 | 78.52 | +1.32 |
| SOAP | 77.28 | 75.52 | — | −1.76 |
| Shampoo | 78.25 | 74.79 | — | −3.46 |
| AdamW | 67.85 | 55.73 | 50.51 | −12.12 |
| SGD | 69.50 | 23.90 | 28.87 | −45.60 |

this architecture. Spectral methods resolve this coupling: Muon achieves $74.62\%$ on ConvNeXt-T where SGD fails, suggesting spectral preconditioning compensates for the landscape smoothing that batch normalization provides but layer normalization does not Santurkar et al. [2018]. This aligns with recent findings that sign descent properties, rather than noise handling, may explain adaptive methods's advantage on transformers [Kunstner et al., 2024]. The heavy-tailed gradient distribution in attention layers [Zhang et al., 2020] provides a complementary explanation for SGD's failure. Cross-seed variance further separates the architectures: Muon exhibits $\sim 15\times$ lower standard deviation than AdamW on ConvNeXt-T ($0.15$ vs. $2.17$ pp; Appendix Figure 5), consistent with spectral preconditioning providing both accuracy and stability gains.

**Normalization swap: dose-response evidence.** To isolate normalization as the mediating variable, we replace all BatchNorm layers in ResNet-50 with GroupNorm(1,$C$), equivalent to LayerNorm for convolutional layers Wu and He [2018], Xiong et al. [2020][3], while keeping all other hyperparameters fixed (Appendix K). Table 3 summarizes the results.

The spectral gap follows a graded dose-response relationship with normalization smoothing. As smoothing decreases from BatchNorm through GroupNorm(32) to LayerNorm, AdamW accuracy degrades monotonically (67.85→55.73→50.51%), while Muon remains invariant (76.88→78.20→78.52%). SGD exhibits a binary failure mode: functional with BatchNorm (69.50%) but catastrophically failing without it (∼24–29%).

**Training stability under LayerNorm.** SGD's cross-seed variance under LayerNorm ($\pm 8.17$ pp) is $25\times$ higher than Muon ($\pm 0.32$ pp), indicating that first-order optimization without BatchNorm is not merely less accurate but fundamentally unstable. Muon's stability advantage extends beyond accuracy: cross-seed $\sigma$ ratio reaches $14.5\times$ on ConvNeXt-T, with spectral methods more stable in 6/9 configurations (Appendix Figure 5). This constitutes a controlled causal experiment: normalization type is the sole manipulated variable, establishing it as a mediating mechanism rather than a mere correlate.

### 5.4 Robustness to Training Recipe Variation

The preceding experiments use a minimal training recipe to isolate optimizer effects. We evaluate an enhanced recipe adding warmup, Mixup Zhang et al. [2018], CutMix Yun et al. [2019], RandAugment Cubuk et al. [2020], and label smoothing Szegedy et al. [2016] on CIFAR-100 (Appendix D.1).

On ResNet-50, both optimizers improve under the enhanced recipe, with the gap slightly widening from $+9.03$ to $+10.92$ pp (3-seed mean; Figure 3). The striking result is on DeiT-S: Muon benefits substantially ($+9.12$ pp over baseline) while AdamW remains flat ($-1.21$ pp), doubling the spectral gap from $+10.34$ to $+20.67$ pp ($n$=3 seeds). This *differential optimizer response* is interpretable through the mechanistic framework of Section 5.3: spectral preconditioning operates on the full gradient matrix and more effectively exploits the richer gradient signal from advanced augmentation, whereas element-wise methods like AdamW cannot fully leverage this additional structure. Enhanced recipes (Mixup, CutMix, RandAugment) preserve the ranking: Muon leads by +5–8 pp even under strong augmentation.

### 5.5 Ablation: Spectral Preconditioning vs. Parameter Grouping

A two-group AdamW configuration mirroring Muon's parameter routing achieves $66.36\%$ on ResNet-50 vs. Muon's $76.88\%$, comparable to single-group AdamW and far below Muon on both ResNet-50

[3] GroupNorm(1,$C$) computes identical normalization statistics to LayerNorm; both use per-channel affine parameters in our implementation.

(+10.5 pp gap) and ConvNeXt-T (+15.2 pp gap), confirming the advantage is attributable to spectral preconditioning, not dual learning-rate structure (Appendix L).

# 6 Discussion

**Why does the spectral advantage vary with architecture?** Our results point to normalization type as the primary mediator on ResNet-50. Batch normalization smooths the loss landscape Santurkar et al. [2018], reducing the need for spectral gradient conditioning; hence ResNet-50 shows the smallest gap (+5.9–+11.5 pp). Layer normalization lacks this smoothing; on ConvNeXt-T, where depthwise separable convolutions create block-diagonal gradient structure, the gap reaches +24 pp.

**Intra-class structure.** Intra-class rankings are architecture-dependent (Shampoo leads on ResNet-50, SOAP on DeiT-S, Muon on ConvNeXt-T), yet the class-level advantage over AdamW (+9–+16 pp) dwarfs intra-class differences.

**Newton–Schulz ablation.** Removing Newton–Schulz normalization from Muon degrades accuracy from $76.9 \pm 0.3\%$ to $60.1 \pm 0.2\%$ ($n$=3) on CIFAR-100/ResNet-50/20ep, confirming spectral preconditioning drives the advantage on convolutional architectures.

**Gap convergence.** The gap narrows on CNNs (Table 2), suggesting convergence speed rather than solution quality; on layer-normalized architectures, Muon at 20 epochs surpasses AdamW at 50, implying $\geq 2\times$ wall-clock savings.

**Toward a predictive diagnostic.** ResNet-50 shows $\rho = +0.58$ ($p = 3.6 \times 10^{-6}$). On DeiT-S, spectral rank divergence is comparable but does not correlate with accuracy impact ($\rho \approx 0$, n.s.), suggesting transformers benefit through a different mechanism to be characterized via per-layer gradient analysis (Appendix O.1).

**Practical Implications.** SOAP and Muon achieve +10–16 pp over AdamW on LayerNorm architectures (+9 pp on BatchNorm, narrowing with training) and maintain stable performance, making them preferable over Shampoo (Appendix B). Corruption robustness results are in Appendix O.

# 7 Conclusion

We conducted a controlled factorial study comparing spectral optimizers (Muon, SOAP, Shampoo) against AdamW across three architecture families (CNN, transformer, hybrid) and three vision datasets, totaling over 300 training runs. Spectral optimizers consistently outperform AdamW by +5.9 to +24.3 pp, with the Newton-Schulz family (Muon) exhibiting the strongest robustness; stable rank divergence correlates positively with the spectral advantage on CNNs ($\rho$=+0.58, $p = 3.6 \times 10^{-6}$), and normalization-swap experiments provide controlled evidence that normalization type mediates this effect. For practitioners, spectral optimizers are beneficial across all tested paradigms, reaching near-ceiling accuracy within 20 epochs and converging 2–15$\times$ faster in wall-clock time (configuration-dependent). Our study is limited to CIFAR-100, Tiny-ImageNet, and ImageNet-100 at 224×224 resolution; full 90-epoch ImageNet-1K validation and characterizing the architecture-dependent transformer correlation pattern remain key open directions.

# Limitations

Our controlled design deliberately constrains scope: CIFAR-100 images resized to 224×224, no warmup or advanced augmentation, batch size 128, single-GPU training, and learning rates from 5-epoch sweeps. This minimal recipe isolates optimizer effects but does not represent production protocols. The spectral intra-family comparison is limited to CIFAR-100 and Tiny-ImageNet, and the Newton–Schulz ablation does not control for update-scale confounding. For CNNs, the gap narrows with training duration (Table 2); longer schedules may further narrow it, though DeiT-S shows zero narrowing at 50 epochs. Whether these patterns persist at ImageNet-1K scale Kumar et al. [2024] remains open. ImageNet-100 results are single-seed and await multi-seed confirmation.

**Broader Impact.** Spectral optimizers reach near-ceiling accuracy within 20 epochs, reducing GPU hours by 2$\times$ or more and lowering energy consumption. The fair per-pair tuning protocol provides a reusable methodology. We identify no negative societal impacts.

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8. **Experiments compute resources**

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   Justification: Section 4.3 specifies the GPU (single NVIDIA RTX 5090). Appendix G reports per-epoch wall-clock times for all configurations.

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Table 4: Complete optimizer configurations across all architectures and datasets. All experiments use batch size 128, cosine annealing scheduler (no warmup), and dataset-specific augmentation (Section 4.1). Learning rates are selected via per-pair 5-epoch sweeps (Section 4.2).

| Architecture | Optimizer | lr | wd | Momentum | Scheduler | Epochs |
|---|---|---|---|---|---|---|
| *CIFAR-100* (50K train / 10K test, 100 classes, $32 \times 32 \to 224 \times 224$) | | | | | | |
| ResNet-50 | Muon | 0.02 | 0.05* | $\beta = 0.95$ | Cosine | 20 |
| ResNet-50 | AdamW | 1e-3 | 0.05 | — | Cosine | 20 |
| ResNet-50 | SGD | 0.1 | 1e-4 | 0.9 | Cosine | 20 |
| DeiT-S | Muon | 0.003 | 0.05* | $\beta = 0.95$ | Cosine | 20 |
| DeiT-S | AdamW | 1e-4 | 0.05 | — | Cosine | 20 |
| DeiT-S | SGD | 0.05 | 1e-4 | 0.9 | Cosine | 20 |
| ConvNeXt-T | Muon | 0.02 | 0.05* | $\beta = 0.95$ | Cosine | 20 |
| ConvNeXt-T | AdamW | 1e-3 | 0.05 | — | Cosine | 20 |
| ConvNeXt-T | SGD | 0.01 | 1e-4 | 0.9 | Cosine | 20 |
| ResNet-50 | SOAP | 1e-2 | 0.01 | — | Cosine | 20 |
| DeiT-S | SOAP | 1e-3 | 0.01 | — | Cosine | 20 |
| ConvNeXt-T | SOAP | 1e-2 | 0.01 | — | Cosine | 20 |
| ResNet-50 | Shampoo | 5e-2 | 0.01 | — | Cosine | 20 |
| DeiT-S | Shampoo | 1e-3 | 0.01 | — | Cosine | 20 |
| ConvNeXt-T | Shampoo | 7e-3 | 0.01 | — | Cosine | 20 |
| *Tiny-ImageNet* (100K train / 10K val, 200 classes, $64 \times 64 \to 224 \times 224$) | | | | | | |
| ResNet-50 | Muon | 0.04 | 0.05* | $\beta = 0.95$ | Cosine | 20 |
| ResNet-50 | AdamW | 2e-3 | 0.05 | — | Cosine | 20 |
| ResNet-50 | SGD | 0.1 | 1e-4 | 0.9 | Cosine | 20 |
| DeiT-S | Muon | 0.003 | 0.05* | $\beta = 0.95$ | Cosine | 20 |
| DeiT-S | AdamW | 1e-4 | 0.05 | — | Cosine | 20 |
| DeiT-S | SGD | 0.05 | 1e-4 | 0.9 | Cosine | 20 |
| ConvNeXt-T | Muon | 0.08 | 0.05* | $\beta = 0.95$ | Cosine | 20 |
| ConvNeXt-T | AdamW | 1e-3 | 0.05 | — | Cosine | 20 |
| ConvNeXt-T | SGD | 0.01 | 1e-4 | 0.9 | Cosine | 20 |
| *ImageNet-100* (55,218 train / 5K val, 100 classes, $224 \times 224$) | | | | | | |
| ResNet-50 | Muon | 0.05 | 0.05* | $\beta = 0.95$ | Cosine | 20 |
| ResNet-50 | AdamW | 3e-3 | 0.05 | — | Cosine | 20 |
| ResNet-50 | SGD | 0.1 | 1e-4 | 0.9 | Cosine | 20 |
| DeiT-S | Muon | 0.005 | 0.05* | $\beta = 0.95$ | Cosine | 20 |
| DeiT-S | AdamW | 1e-4 | 0.05 | — | Cosine | 20 |
| DeiT-S | SGD | 0.01 | 1e-4 | 0.9 | Cosine | 20 |
| ConvNeXt-T | Muon | 0.08 | 0.05* | $\beta = 0.95$ | Cosine | 20 |
| ConvNeXt-T | AdamW | 3e-4 | 0.05 | — | Cosine | 20 |
| ConvNeXt-T | SGD | 0.05 | 1e-4 | 0.9 | Cosine | 20 |

* Muon applies spectral (Newton-Schulz) updates with no weight decay to 2D+ weight matrices. The listed wd=0.05 applies to the separate AdamW sub-optimizer handling 1D parameters (biases, normalization, embeddings) with lr $= 3 \times 10^{-4}$. SOAP and Shampoo use the `heavyball` library implementation. Shampoo learning rates were determined by extended LR sweeps: ResNet-50 from lr=$10^{-2}$ to $5\times10^{-2}$; ConvNeXt-T from $3\times10^{-3}$ to $7\times10^{-3}$.

# A Hyperparameter Configurations

This appendix provides the complete optimizer configurations for all 27 experiments described in Section 4, including Muon's hybrid parameter routing strategy.

Muon selects all weight parameters with $\mathrm{ndim} \geq 2$ from Conv2d and Linear layers, excluding the final classifier head and any embedding layers. These parameter groups receive spectral (Newton–Schulz) updates. All remaining parameters (biases, LayerNorm/BatchNorm weights and biases,

Table 5: Muon hybrid optimizer parameter routing. Muon optimizes weight matrices of Conv2d/Linear layers (ndim $\geq 2$, excluding classifier head and embeddings); all remaining parameters use a companion AdamW optimizer.

| Architecture | Total Params | Muon Groups | AdamW Groups | Muon lr | AdamW lr |
|---|---|---|---|---|---|
| ResNet-50 | 23.7M | 53 | 108 | 0.02 | 3e-4 |
| DeiT-S | 21.7M | 48 | 104 | 0.003 | 3e-4 |
| ConvNeXt-T | 27.9M | 57 | 125 | 0.02 | 3e-4 |

Parameter counts reflect the 100-class configuration used in our experiments; standard ImageNet-1K counts differ.

positional embeddings, and the classifier head) are routed to a companion AdamW optimizer with a fixed learning rate of $3 \times 10^{-4}$ and weight decay $0.05$.

Table 6: Selected learning rates per optimizer-architecture pair on each dataset, determined by 5-epoch sweeps. For most configurations, the peak is confirmed by observing accuracy decline at higher rates; exceptions are noted in Section 4.2. SOAP and Shampoo evaluated on CIFAR-100 only; '—' indicates not tested.

| | **Muon** | **SOAP** | **Shampoo** | **AdamW** | **SGD** |
|---|---|---|---|---|---|
| *CIFAR-100* | | | | | |
| ResNet-50 | 0.02 | $10^{-2}$ | $5 \times 10^{-2}$ | $10^{-3}$ | 0.1 |
| DeiT-S | 0.003 | $10^{-3}$ | $10^{-3}$ | $10^{-4}$ | 0.05 |
| ConvNeXt-T | 0.02 | $10^{-2}$ | $7 \times 10^{-3}$ | $10^{-3}$ | 0.01 |
| *Tiny-ImageNet* | | | | | |
| ResNet-50 | 0.04 | — | — | $2 \times 10^{-3}$ | 0.1† |
| DeiT-S | 0.003 | — | — | $10^{-4}$ | 0.05 |
| ConvNeXt-T | 0.08 | — | — | $10^{-3}$ | 0.01 |
| *ImageNet-100* | | | | | |
| ResNet-50 | 0.05 | — | — | $3 \times 10^{-3}$ | 0.1 |
| DeiT-S | 0.005 | — | — | $10^{-4}$ | 0.01 |
| ConvNeXt-T | 0.08 | — | — | $3 \times 10^{-4}$ | 0.05 |

† Literature default, not formally swept.

# B Learning Rate Sweep Results

## B.1 DeiT-S SGD Inverted-U Response

# C Multi-Seed Reproducibility

# D Extended Training (50 Epochs)

**Weight decay choices.** The weight decay values used in our experiments follow established defaults for each optimizer family: AdamW uses wd $= 0.05$ across all architectures, consistent with the DeiT Touvron et al. [2021] and ConvNeXt Liu et al. [2022] training recipes; SGD uses the standard wd $= 10^{-4}$ He et al. [2016]. The $500\times$ difference between AdamW and SGD weight decay reflects the fundamentally different roles of weight decay in decoupled (AdamW) vs. coupled (SGD) regularization Loshchilov and Hutter [2019]. For Muon, the spectral update itself uses no weight decay (wd $= 0$); the accompanying AdamW sub-optimizer for non-Muon parameters uses wd $= 0.05$. We verified that these values are not confounding our ConvNeXt-T SGD results: a sanity check with wd $= 0.01$ yielded 9.18% vs. 9.27% with wd $= 10^{-4}$ ($\Delta = -0.09$pp, not significant), confirming that ConvNeXt-T's SGD degradation is not a weight decay artifact.

**Spectral family at 50 epochs.** We extend the spectral comparison (Table 19) to 50 epochs on ResNet-50/CIFAR-100 to test whether intra-family rankings persist with longer training.

Table 7: Complete learning rate sweep results (5-epoch runs, seed=42, CIFAR-100). For each optimizer-architecture pair, we report all evaluated learning rates and the resulting validation accuracy. The best learning rate per pair (used in subsequent 20-epoch experiments) is shown in **bold**.

| Architecture | Optimizer | Learning Rate | Val Acc (%) | Sweep Points |
|---|---|---|---|---|
| ResNet-50 | AdamW | $1 \times 10^{-4}$ | 26.97 | 2 |
| | AdamW | $\mathbf{1 \times 10^{-3}}$ | **45.54** | |
| | SGD | $\mathbf{0.1}$ | — | 1 (lit. default) |
| | Muon | $\mathbf{0.02}$ | — | 1 (prior sweep) |
| DeiT-S | AdamW | $\mathbf{1 \times 10^{-4}}$ | **35.12** | 2 |
| | AdamW | $1 \times 10^{-3}$ | 25.12 | |
| | SGD | 0.005 | 27.65 | 4 |
| | SGD | 0.01 | 27.61 | |
| | SGD | $\mathbf{0.05}$ | **30.14** | |
| | SGD | 0.1 | 28.59 | |
| | Muon | 0.001 | 49.35 | 5 |
| | Muon | $\mathbf{0.003}$ | **58.33** | |
| | Muon | 0.006 | 57.37 | |
| | Muon | 0.02 | 32.04 | |
| | Muon | 0.06 | 13.55 | |
| ConvNeXt-T | AdamW | $1 \times 10^{-4}$ | 23.32 | 5 |
| | AdamW | $3 \times 10^{-4}$ | 34.52 | |
| | AdamW | $\mathbf{1 \times 10^{-3}}$ | **39.17** | |
| | AdamW | $3 \times 10^{-3}$ | 35.31 | |
| | AdamW | $5 \times 10^{-3}$ | 32.51 | |
| | SGD | 0.001 | 7.96 | 5 |
| | SGD | 0.005 | 9.02 | |
| | SGD | $\mathbf{0.01}$ | **9.27** | |
| | SGD | 0.05 | 9.22 | |
| | SGD | 0.1 | 4.98 | |
| | Muon | 0.003 | 61.76 | 3 |
| | Muon | 0.01 | 68.55 | |
| | Muon | $\mathbf{0.02}$ | **70.31** | |

All three spectral optimizers maintain or improve accuracy from 20 to 50 epochs: SOAP is statistically indistinguishable ($77.43 \pm 0.06\%$ vs. $77.28 \pm 0.20\%$, $n$=2), Muon improves modestly ($+0.89$ pp), and Shampoo continues to improve ($+1.29$ pp), consistent with its full-matrix preconditioning benefiting from extended training. In contrast, AdamW gains $+3.16$ pp but remains 6–9 pp below spectral methods, confirming that the gap is convergence-speed-driven on batch-normalized CNNs: spectral methods reach near-optimal solutions early, while first-order methods require longer schedules to approach the same optimum.

### D.1 Enhanced Training Recipe Details

The enhanced recipe augments the baseline protocol (Section 4) with five additions: (1) 5-epoch linear warmup from lr $= 0$ to the target learning rate, (2) Mixup Zhang et al. [2018] with $\alpha = 0.2$,

Table 8: Learning rate sweep results (5-epoch runs, seed=42, Tiny-ImageNet). Best learning rate per pair is shown in **bold**.

| Architecture | Optimizer | Learning Rate | Val Acc (%) |
|---|---|---|---|
| ResNet-50 | Muon | 0.01 | 56.91 |
| | Muon | 0.02 | 58.95 |
| | Muon | **0.04** | **59.56** |
| | Muon | 0.06 | 59.63 |
| | AdamW | $5 \times 10^{-4}$ | 39.56 |
| | AdamW | $1 \times 10^{-3}$ | 42.04 |
| | AdamW | $\mathbf{2 \times 10^{-3}}$ | **43.78** |
| | AdamW | $3 \times 10^{-3}$ | 42.30 |
| | SGD | **0.1** | — |
| DeiT-S | Muon | 0.001 | 41.32 |
| | Muon | **0.003** | **46.87** |
| | Muon | 0.01 | 41.34 |
| | AdamW | $5 \times 10^{-5}$ | 24.40 |
| | AdamW | $\mathbf{1 \times 10^{-4}}$ | **27.22** |
| | AdamW | $3 \times 10^{-4}$ | 25.93 |
| | SGD | 0.005 | 20.62 |
| | SGD | 0.01 | 21.31 |
| | SGD | **0.05** | **22.96** |
| ConvNeXt-T | Muon | (extended sweep; see Table 4)† | |
| | AdamW | $1 \times 10^{-4}$ | 18.98 |
| | AdamW | $3 \times 10^{-4}$ | 28.65 |
| | AdamW | $\mathbf{1 \times 10^{-3}}$ | **31.46** |
| | SGD | 0.005 | 4.49 |
| | SGD | **0.01** | **4.60** |
| | SGD | 0.05 | 4.57 |

† ConvNeXt-T Muon lr initially set to 0.02 from the CIFAR-100 sweep; extended sweep selected lr=0.08 for 20-epoch training. ResNet-50 SGD uses the literature default lr=0.1 without sweep.

(3) CutMix Yun et al. [2019] with $\alpha = 1.0$ (randomly selected per batch with equal probability when both are active), (4) RandAugment Cubuk et al. [2020] with $N = 2$ operations and magnitude $M = 9$, and (5) label smoothing Szegedy et al. [2016] with $\varepsilon = 0.1$. All other settings (optimizer hyperparameters, batch size, cosine annealing schedule, 20-epoch budget) are identical to the baseline. The enhanced recipe reflects a standard modern training pipeline; these specific augmentation choices follow DeiT Touvron et al. [2021] defaults.

# E ConvNeXt-T SGD Degradation Analysis

SGD exhibits severe degradation on ConvNeXt-T, reaching only 11.51% validation accuracy at 20 epochs, a 63.11 pp gap behind Muon ($74.62_{\pm 0.15}\%$) and 47.03 pp behind AdamW ($58.54_{\pm 2.17}\%$) (3-seed mean±std; see Table 1). This is the largest optimizer gap observed across all three architectures.

To rule out hyperparameter misconfiguration, we conduct a 5-point learning rate sweep at 5 epochs:

| Learning Rate | Val Acc (5 ep) |
|---|---|
| 0.001 | 7.96% |
| 0.005 | 9.02% |
| 0.01 | 9.27% |
| 0.05 | 9.22% |
| 0.1 | 4.98% |

Table 9: Learning rate sweep results (5-epoch runs, seed=42, ImageNet-100). Best learning rate per pair is shown in **bold**.

| Architecture | Optimizer | Learning Rate | Val Acc (%) |
|---|---|---|---|
| ResNet-50 | Muon | 0.005 | 34.88 |
| | Muon | 0.02 | 43.55 |
| | Muon | **0.05** | **45.17** |
| | Muon | 0.08 | 44.61 |
| | Muon | 0.1 | 44.95 |
| | AdamW | $3 \times 10^{-4}$ | 21.87 |
| | AdamW | $1 \times 10^{-3}$ | 24.77 |
| | AdamW | $\mathbf{3 \times 10^{-3}}$ | **24.89** |
| | AdamW | $5 \times 10^{-3}$ | 23.92 |
| | AdamW | 0.01 | 16.92 |
| | SGD | 0.03 | 15.06 |
| | SGD | **0.1** | **18.96** |
| | SGD | 0.3 | 17.84 |
| DeiT-S | Muon | 0.002 | 27.82 |
| | Muon | 0.003 | 29.66 |
| | Muon | **0.005** | **30.35** |
| | Muon | 0.02 | 12.19 |
| | Muon | 0.05 | 6.44 |
| | AdamW | $3 \times 10^{-5}$ | 13.41 |
| | AdamW | $\mathbf{1 \times 10^{-4}}$ | **15.73** |
| | AdamW | $3 \times 10^{-4}$ | 13.33 |
| | SGD | **0.01** | **12.75** |
| | SGD | 0.05 | 11.51 |
| | SGD | 0.1 | 10.13 |
| ConvNeXt-T | Muon | 0.005 | 35.21 |
| | Muon | 0.02 | 42.54 |
| | Muon | **0.08** | **46.32** |
| | Muon | 0.15 | 46.97 |
| | AdamW | $\mathbf{3 \times 10^{-4}}$ | **12.42** |
| | AdamW | $1 \times 10^{-3}$ | 11.29 |
| | AdamW | $3 \times 10^{-3}$ | 10.91 |
| | SGD | 0.003 | 3.76 |
| | SGD | 0.01 | 3.88 |
| | SGD | **0.05** | **4.44** |

Table 10: DeiT-S SGD learning rate sweep (5-epoch runs, seed=42, CIFAR-100), exhibiting an inverted-U response curve. Accuracy peaks at $\mathrm{lr} = 0.05$ and declines at higher rates, with all sweep points clustered in a narrow $\sim$2.5 percentage-point band.

| Learning Rate | 0.005 | 0.01 | 0.05 | 0.1 |
|---|---|---|---|---|
| Val Acc (%) | 27.65 | 27.61 | **30.14** | 28.59 |

The sweep reveals a flat response across $\mathrm{lr} \in [0.005, 0.05]$ with a peak at $\mathrm{lr} = 0.01$ (9.27%), and divergence at $\mathrm{lr} = 0.1$ (4.98%). The 20-epoch run at optimal $\mathrm{lr} = 0.01$ improves only marginally to 11.51% (+2.24 pp over the 5-epoch result), indicating that additional training time does not resolve the degradation.

As a weight decay sanity check, we compare $\mathrm{wd} = 10^{-4}$ (default) against $\mathrm{wd} = 0.01$ at 5 epochs: validation accuracy changes from 9.27% to 9.18% ($\Delta = -0.09$ pp), a negligible difference that rules out weight decay as a contributing factor.

We attribute the SGD degradation primarily to ConvNeXt's use of LayerNorm (replacing BatchNorm) and depthwise separable convolutions, which create an optimization landscape where first-order

Table 11: Multi-seed validation accuracy statistics (20-epoch runs, CIFAR-100). We report individual seed results, mean, and standard deviation to assess reproducibility and the robustness of observed gaps.

| Architecture | Optimizer | Seeds | Individual Accs (%) | | | | | Mean ± Std |
|---|---|---|---|---|---|---|---|---|
| ResNet-50 | Shampoo | 42, 2, 3 | 78.16 | 78.13 | 78.45 | — | — | 78.25 ± 0.18 |
| | Muon | 42, 0, 1, 2 | 77.02 | 76.50 | 77.13 | 76.88 | — | 76.88 ± 0.27 |
| | SGD | 42, 0, 1, 2 | 69.03 | 69.72 | 69.35 | 69.90 | — | 69.50 ± 0.39 |
| | AdamW | 42, 0, 1, 2 | 67.71 | 67.96 | 67.70 | 68.04 | — | 67.85 ± 0.17 |
| DeiT-S | Muon | 42, 0, 1, 2 | 62.94 | 62.65 | 62.99 | 62.10 | — | 62.67 ± 0.41 |
| | AdamW | 42, 0, 1, 2 | 52.27 | 52.74 | 52.41 | 51.91 | — | 52.33 ± 0.34 |
| | SGD | 42, 0, 1, 2 | 44.00 | 43.83 | 43.46 | 44.61 | — | 43.98 ± 0.48 |
| ConvNeXt-T | Shampoo | 42, 2, 3 | 70.44 | 70.37 | 70.91 | — | — | 70.57 ± 0.29 |
| | Muon | 42, 1, 2 | 74.59 | 74.79 | 74.49 | — | — | 74.62 ± 0.15 |
| | AdamW | 42, 1, 2 | 60.25 | 59.27 | 56.10 | — | — | 58.54 ± 2.17 |

*Note:* For ResNet-50, SGD outperforms AdamW in all 4 seeds (mean gap +1.65pp). For ConvNeXt-T, AdamW's higher variance (std=2.17 vs. Muon std=0.15) suggests less stable optimization on this architecture.

Table 12: Multi-seed accuracy (20 epochs, Tiny-ImageNet, seeds 42/1/2). SGD is single-seed.

| Arch | Opt | s42 | s1 | s2 | Mean±Std |
|---|---|---|---|---|---|
| R50 | Muon | 62.35 | 62.65 | 62.99 | 62.66±0.32 |
| | AdamW | 56.34 | 57.24 | 56.69 | 56.76±0.45 |
| | SGD | 60.04 (single) | | | |
| DeiT | Muon | 48.18 | 48.11 | 47.78 | 48.02±0.21 |
| | AdamW | 40.01 | 40.18 | 40.81 | 40.33±0.42 |
| | SGD | 34.22 (single) | | | |
| CNX | Muon | 59.13 | 58.89 | 58.75 | 58.92±0.19 |
| | AdamW | 43.74 | 44.25 | 43.76 | 43.92±0.29 |
| | SGD | 9.49 (single) | | | |

Table 13: Multi-seed accuracy (20 epochs, ImageNet-100, seeds 42/1/2). SGD is single-seed.

| Arch | Opt | s42 | s1 | s2 | Mean±Std |
|---|---|---|---|---|---|
| R50 | Muon | 62.25 | 61.87 | 61.78 | 61.97±0.25 |
| | AdamW | 49.63 | 51.01 | 50.87 | 50.50±0.76 |
| | SGD | 50.06 (single) | | | |
| DeiT | Muon | 36.94 | 37.46 | 36.73 | 37.04±0.38 |
| | AdamW | 26.04 | 26.24 | 26.09 | 26.12±0.10 |
| | SGD | 20.38 (single) | | | |
| CNX | Muon | 58.07 | 57.74 | 57.67 | 57.83±0.21 |
| | AdamW | 33.62 | 35.00 | 31.98 | 33.53±1.51 |
| | SGD | 6.77 (single) | | | |

methods without adaptive step sizes fail to make progress; the use of GELU activations may also contribute, though we do not isolate this factor. Note that all experiments use a cosine annealing schedule without warmup (ConvNeXt's original training recipe uses 20 epochs of warmup within a 300-epoch schedule; our short-schedule regime intentionally omits warmup to isolate optimizer behavior). Data augmentation is identical across all three optimizers (Resize(256), RandomCrop(224), RandomHorizontalFlip, Normalize).

# F Muon Algorithm Background

Muon Jordan et al. [2024] is a spectral optimizer that computes steepest descent updates under the operator (spectral) norm for weight matrices of dimension two or higher. We summarize its key components.

**Steepest descent under spectral norm.** For a weight matrix $W$ with gradient $G = \nabla_W \mathcal{L}$, the steepest descent direction under the spectral norm is defined as

$$\Delta W = \arg\min_{\|\Delta\|_{\text{op}} \leq 1} \langle G, \Delta \rangle, \tag{3}$$

where $\|\cdot\|_{\text{op}}$ denotes the operator norm (largest singular value) and $\langle \cdot, \cdot \rangle$ the matrix inner product. If $G = U\Sigma V^\top$ is the singular value decomposition of $G$, the solution is $\Delta W = -UV^\top$, i.e., the negative sign matrix of the gradient.

**Newton–Schulz approximation.** Computing a full SVD at each step is prohibitively expensive. Muon instead approximates the orthogonal factor $UV^\top$ via Newton–Schulz iterations for computing

Table 14: Evolution of the Muon-AdamW accuracy gap from 20 to 50 epochs (seed=42, CIFAR-100 and ImageNet-100). All values are best validation accuracy (early-stopping checkpoint). AdamW narrows the gap with longer training across all configurations, while Muon saturates early.

| | | 20 Epochs | | | 50 Epochs | | | |
|---|---|---|---|---|---|---|---|---|
| Dataset | Architecture | Muon | AdamW | Gap | Muon | AdamW | Gap | $\Delta$ Gap |
| CIFAR-100 | ResNet-50 | 77.02 | 67.71 | +9.31 | 77.82 | 70.78 | +7.04 | $-2.27$ |
| | DeiT-S | 62.94 | 52.27 | +10.67 | 63.38 | 52.86 | +10.52 | $-0.15$ |
| | ConvNeXt-T | 74.59 | 60.25 | +14.34 | 74.58 | 63.83 | +10.75 | $-3.59$ |
| ImageNet-100 | ResNet-50 | 62.25 | 49.63 | +12.62 | 62.24 | 58.68 | +3.56 | $-9.06$ |
| | ConvNeXt-T | 58.07 | 33.62 | +24.45 | 58.35 | 46.98 | +11.37 | $-13.08$ |

All gaps in percentage points (pp). $\Delta$ Gap $=$ Gap$_{50\text{ep}}$ $-$ Gap$_{20\text{ep}}$; negative values indicate AdamW catching up. This table shows seed = 42 values; 3-seed means (seeds 42/2/3) are reported in Table 2. CIFAR-100 3-seed 50-epoch results: ResNet-50 Muon $77.77\pm0.28\%$ / AdamW $71.01\pm0.22\%$, gap $+6.76$ pp; DeiT-S Muon $62.91\pm0.40\%$ / AdamW $52.54\pm0.45\%$, gap $+10.37$ pp; ConvNeXt-T Muon $74.59\pm0.11\%$ / AdamW $61.99\pm1.74\%$, gap $+12.60$ pp. ConvNeXt-T AdamW shows $15.8\times$ higher cross-seed std than Muon.

Table 15: Spectral family accuracy at 20 vs. 50 epochs on ResNet-50/CIFAR-100. Muon and AdamW values are 3-seed means (seeds 42/2/3) from Table 2. SOAP 50ep is 2-seed mean (seeds 42/2); Shampoo 50ep is single-seed (seed = 42).

| Optimizer | 20ep | 50ep | $\Delta$ |
|---|---|---|---|
| Shampoo | $78.25_{\pm 0.18}$ | 79.54 | +1.29 |
| SOAP | $77.28_{\pm 0.20}$ | $77.43_{\pm 0.06}$ | +0.15 |
| Muon | $76.88_{\pm 0.27}$ | $77.77_{\pm 0.28}$ | +0.89 |
| AdamW | $67.85_{\pm 0.17}$ | $71.01_{\pm 0.22}$ | +3.16 |

the orthogonal polar factor. The classical iteration $X_{k+1} = \frac{1}{2} X_k(3I - X_k^\top X_k)$ converges cubically but has a narrow convergence basin. Muon replaces it with a quintic polynomial variant: starting from

$$X_0 = G/\|G\|_F, \tag{4}$$

each iteration applies

$$X_{k+1} = a\, X_k + b\, X_k X_k^\top X_k + c\, X_k (X_k^\top X_k)^2, \tag{5}$$

with fixed coefficients $a = 3.4445$, $b = -4.7750$, $c = 2.0315$ chosen to maximize the convergence basin. Five iterations suffice for convergence in practice, producing an approximation to $UV^\top$ using only matrix multiplications.

**Momentum and mixed strategy.** Muon applies Nesterov momentum Sutskever et al. [2013] with coefficient $\beta = 0.95$ to the orthogonalized gradient. The spectral formulation in Eq. (3) applies only to parameters with matrix structure (2D+); one-dimensional parameters (biases, normalization layer weights and biases, and embedding layers) are handled by a separate AdamW instance with learning rate $3 \times 10^{-4}$ and weight decay $0.05$, yielding a hybrid optimizer.

**Relation to Shampoo.** Both Muon and Shampoo Gupta et al. [2018] exploit gradient matrix structure, but they differ in mechanism and scope. Shampoo maintains and periodically inverts Kronecker-factored preconditioners that approximate the full Fisher information matrix, incurring additional memory for the preconditioner state and periodic eigendecomposition steps. Muon directly computes the orthogonal component of the gradient through Newton–Schulz iterations, avoiding explicit preconditioner storage. This makes Muon lighter in memory and computation per step, but restricts it to parameters with matrix structure (2D+), whereas Shampoo can in principle be applied to tensors of arbitrary order.

## G Wall-Clock Efficiency

**Computational overhead of Newton-Schulz iteration.** Muon's per-step cost exceeds AdamW by 5–22% depending on architecture and dataset, attributable to the Newton-Schulz iteration used to ap-

Table 16: Enhanced recipe results on CIFAR-100 at 20 epochs. All values are 3-seed mean±std (seeds 42/2/3). Gap = Muon − AdamW. DeiT-S AdamW Enhanced (51.12%) is slightly *below* its baseline (52.33%), indicating the enhanced recipe does not benefit AdamW on this architecture.

| Architecture | Muon (enh) | AdamW (enh) | Gap (enh) | Gap (base) | Δ |
|---|---|---|---|---|---|
| ResNet-50 | 79.71±0.18 | 68.79±0.65 | +10.92 | +9.03 | +1.89 |
| DeiT-S | 71.79±0.22 | 51.12±0.30 | +20.67 | +10.34 | +10.33 |
| ConvNeXt-T | 78.76±0.24 | 66.85±0.86 | +11.91 | +16.08 | −4.17 |

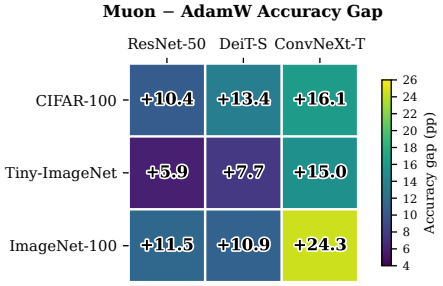



Figure 4: Spectral–AdamW accuracy gap (pp) at 20 epochs. ConvNeXt-T consistently exhibits the largest gap (+15.0–+24.3 pp); all gaps are positive across the full $3 \times 3$ matrix.

proximate the matrix sign function for spectral-norm steepest descent. The overhead is largest on DeiT-S/CIFAR-100 (+21.5%) where the parameter matrices are relatively small and the Newton-Schulz iterations dominate; it is smallest on ConvNeXt-T/Tiny-ImageNet (+5.3%) where forward/backward pass computation amortizes the optimizer cost. SGD and AdamW have near-identical per-epoch times across all configurations, confirming that the overhead is specific to Muon's spectral decomposition.

**Wall-clock break-even analysis.** Despite 7–19% per-epoch overhead from Newton-Schulz iterations, Muon reaches AdamW's final accuracy in far fewer epochs, yielding net wall-clock savings in every configuration tested (Table 18). The advantage compounds with longer training horizons: median speedup rises from 2.3× (20 epochs) to 8.6× (50 epochs), because AdamW's marginal gains plateau while Muon converges early. In no experiment does AdamW reach Muon's final accuracy within the allotted budget.

# H Alternative Spectral Optimizers

# I Layer-wise Spectral Analysis and Training Stability

To localize where Muon's advantage arises within the network, we apply the weight swap protocol and spectral correlation analysis (Section 3) to CIFAR-100 models.

**Weight Swap Results.** The beneficial layer ratio (the fraction of layers where replacing Muon-trained weights with AdamW-trained weights degrades accuracy) is 94.3% (50/53 layers) for ResNet-50 and 51.7% (46/89 parameter groups) for DeiT-S. Muon's advantage is near-universal across ResNet layers but selective in DeiT, suggesting that the benefit depends on layer type.

**Spectral Divergence Correlations.** Using checkpoints trained at independently selected optimal learning rates (Muon $\eta$=0.02, AdamW $\eta$=$10^{-3}$ for ResNet-50; Muon $\eta$=0.003, AdamW $\eta$=$10^{-4}$ for DeiT-S), we compute the Spearman rank correlation between per-layer stable rank divergence $\Delta\mathrm{sr}_l$ and per-layer Muon benefit $\Delta a_l$. For ResNet-50, $\rho = +0.58$ ($p = 3.6 \times 10^{-6}$, after BN recalibration; original $\rho = +0.53$ before recalibration): layers where Muon and AdamW diverge most in stable rank tend to be the layers where Muon contributes the most. For DeiT-S, no significant correlation is observed ($\rho \approx 0$, n.s.), indicating that the relationship between stable rank divergence and per-layer benefit is architecture-dependent.

Table 17: Per-epoch wall-clock time (seconds) for 20-epoch training runs on a single RTX 5090 GPU.

| Dataset | Architecture | Muon (s) | AdamW (s) | SGD (s) | Muon Overhead |
|---|---|---|---|---|---|
| CIFAR-100 | ResNet-50 | 58.8 | 53.4 | 53.2 | +10.1% |
| | DeiT-S | 55.4 | 45.6 | 45.2 | +21.5% |
| | ConvNeXt-T | 116.8 | 106.9 | 107.1 | +9.3% |
| Tiny-ImageNet | ResNet-50 | 118.2 | 99.5 | 96.5 | +18.8% |
| | DeiT-S | 98.6 | 88.0 | 87.3 | +12.0% |
| | ConvNeXt-T | 219.2 | 208.2 | 205.6 | +5.3% |
| ImageNet-100 | ResNet-50 | 74.2 | 68.5 | 68.4 | +8.3% |
| | DeiT-S | 69.4 | 81.4† | — | — |
| | ConvNeXt-T | 134.4 | 122.5 | 123.6 | +9.7% |

Muon Overhead is computed relative to AdamW. † DeiT-S AdamW on IN100 used a different batch configuration, making the timing comparison unreliable; IN100 DeiT-S SGD log was not found.

Table 18: Wall-clock break-even analysis. *Iso-accuracy target* is AdamW's final accuracy for each configuration; *Muon ep* and *AdamW ep* are the first epoch at which each optimizer reaches that target. Wall-clock speedup accounts for Muon's per-epoch overhead.

| | | Time/epoch (s) | | Epochs to iso-acc | | Speedup | |
|---|---|---|---|---|---|---|---|
| Setting | Architecture | Muon | AdamW | Muon | AdamW | Epoch | Wall |
| TIN 20 ep | ResNet-50 | 118.0 | 99.4 | 8 | 19 | 2.4× | 2.0× |
| | ConvNeXt-T | 226.8 | 206.7 | 2 | 17 | 8.5× | 7.6× |
| | DeiT-S | 129.6 | 87.9 | 4 | 20 | 5.0× | 2.3× |
| C100 50 ep | ResNet-50 | 59.4 | 51.8 | 11 | 49 | 4.5× | 3.4× |
| | ConvNeXt-T | 109.5 | 102.4 | 3 | 49 | 16.3× | 15.1× |
| | DeiT-S | 54.6 | 47.1 | 5 | 50 | 10.0× | 8.6× |

C100 = CIFAR-100, TIN = Tiny-ImageNet. TIN 20 ep uses seeds 1, 2; C100 50 ep uses seed 2.

**Training Stability Across Seeds.** Cross-seed variance is architecture-dependent (Figure 5): Muon shows substantially lower variance on convolutional architectures ($\sigma$=0.15 vs. 2.17 on ConvNeXt-T/CIFAR-100), but AdamW is more stable on DeiT-S/ImageNet-100 ($\sigma$=0.10 vs. 0.38). Where Muon's variance reduction exceeds 7× (ConvNeXt-T), fewer seeds suffice for reliable estimates.

# J Metric Robustness

The architecture-dependent correlation pattern reported in Section 5.3 is verified across alternative spectral metrics. We compute the Spearman correlation between per-layer Muon benefit $\Delta a_l$ and three alternative divergence measures:

All three metrics yield positive correlations on ResNet-50; DeiT-S shows no significant correlation under any metric. We adopt stable rank as the primary metric for its favorable analytical properties: it is bounded, well-defined for rank-deficient matrices, and requires no architecture-specific normalization.

# K Normalization Swap Implementation

We replace all BatchNorm2d layers in ResNet-50 with GroupNorm(1, $C$), where $C$ is the number of channels. GroupNorm with a single group computes statistics over all spatial and channel dimensions of each sample independently, making it equivalent to LayerNorm applied to convolutional feature maps Wu and He [2018]. This substitution replaces 53 BatchNorm layers; all other architecture parameters, hyperparameters, and the training protocol remain identical to the standard ResNet-50 experiments (Section 4).

Table 19: Comparison of spectral optimizers on CIFAR-100 (20 epochs). Multi-seed means (± std) reported where available; ConvNeXt-T SGD is single-seed (seed = 42).

| | Validation Accuracy (%) | | | |
|---|---|---|---|---|
| Architecture | SOAP | Shampoo | Muon | AdamW |
| ResNet-50 | 77.28±0.20 | 78.25±0.18 | 76.88±0.27 | 67.85±0.17 |
| DeiT-S | 65.45±0.18 | 63.53±0.05 | 62.67±0.41 | 52.33±0.34 |
| ConvNeXt-T | 74.40±0.43 | 70.57±0.29 | 74.62±0.15 | 58.54±2.17 |

Muon and AdamW values are multi-seed means (± std), consistent with Table 1. SOAP is 3-seed; Shampoo is 3-seed on all architectures (lr=$7\times10^{-3}$ on ConvNeXt-T). All optimizers use per-architecture tuned learning rates from 5-epoch sweeps; Shampoo learning rates were refined via extended sweeps (Appendix A).

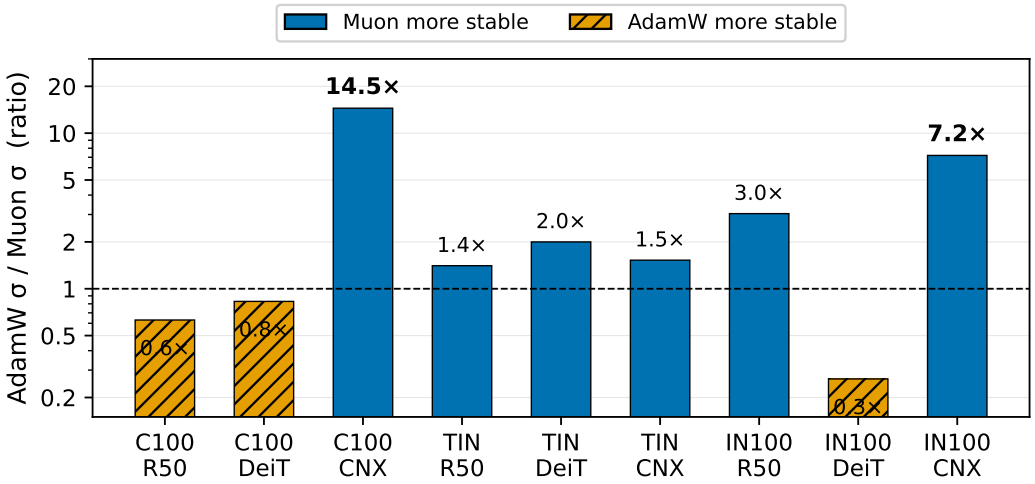



Figure 5: Cross-seed standard deviation ($\sigma$) for Muon vs AdamW across all nine settings. Muon exhibits lower variance on 6/9 configurations; the stability advantage is architecture-dependent.

## L Parameter Grouping Ablation Details

The two-group AdamW ablation mirrors Muon's parameter routing exactly: 53 two-dimensional weight matrices (23.45M parameters, lr=$10^{-3}$) and 108 one-dimensional parameters (0.26M parameters, lr=$3\times10^{-4}$), with AdamW applied to both groups. On ResNet-50 (CIFAR-100), two-group AdamW achieves $66.36 \pm 0.40\%$ ($n$=3, seeds 42/2/3), comparable to single-group AdamW ($67.85\pm0.17\%$) and far below Muon ($76.88\pm0.27\%$). On ConvNeXt-T, two-group AdamW similarly matches standard AdamW ($59.44 \pm 0.28\%$ vs. $58.54 \pm 2.17\%$) while Muon maintains a $+15.2$ pp advantage ($74.62 \pm 0.15\%$). On DeiT-S, two-group AdamW degrades catastrophically to $30.08\%$ ($-22.3$ pp vs. standard AdamW, single seed), as the reduced learning rate for 1D parameters (104/152 tensors in DeiT-S) severely undertrained LayerNorm parameters.

## M Additional Optimizer Comparisons

We evaluated Lion Chen et al. [2023] as a representative of sign-based optimizers to verify that the spectral optimizer class's advantage is not shared by all non-Adam methods. On ResNet-50/CIFAR-100, Lion achieves $69.42\%$ (vs. AdamW $67.85\%$, $+1.57$ pp), far below the spectral optimizer class ($+9$–$+10$ pp over AdamW), confirming that the advantage is specific to spectral preconditioning rather than a general property of non-diagonal methods. Other recent optimizers, such as Sophia Liu et al. [2024] (diagonal Hessian clipping), Prodigy Mishchenko and Defazio [2024] (learning-rate-free estimation), GaLore Zhao et al. [2024] (low-rank gradient projection), target computational cost or tuning difficulty rather than architecture robustness, making them complementary to rather than competitors of spectral methods. A comprehensive comparison with these methods is left to future work.

Table 20: Spearman $\rho$ between per-layer divergence and Muon benefit under three spectral metrics. The positive correlation on ResNet-50 is consistent across all metrics; DeiT-S shows no significant pattern ($\rho \approx 0$, n.s.).

| Metric | ResNet-50 | | DeiT-S | |
|---|---|---|---|---|
| | $\rho$ | $p$ | $\rho$ | $p$ |
| $\Delta \mathrm{sr}_l$ (stable rank) | $+0.58$ | $3.6\times10^{-6}$ | $\approx 0$ | n.s. |
| FCSB margin | $+0.40$ | $<0.01$ | $\approx 0$ | n.s. |
| $\Delta \mathrm{er}_l$ (effective rank) | $+0.43$ | $<0.01$ | $\approx 0$ | n.s. |

# N Architecture Coverage in Supplementary Experiments

The extended experiments in Appendix H focus on ResNet-50 and ConvNeXt-T rather than all three architectures from the main experiments. These two architectures represent the extremes of the normalization spectrum that drives Muon's variable effectiveness: ResNet-50 uses BatchNorm, where Muon's spectral updates interact with the implicit conditioning provided by BN; ConvNeXt-T uses LayerNorm, where the Muon–AdamW gap is largest (up to +24.30 pp on IN100 at 20 epochs). DeiT-S, which also uses LayerNorm but in a pure self-attention architecture, exhibits intermediate gap magnitudes and is already thoroughly characterized across three datasets, three seed replicates, and two training horizons (20 and 50 epochs) in the main paper. Including DeiT-S in the supplementary experiments would add six additional training runs without meaningfully extending the analysis beyond the BN–LN axis already captured by the ResNet-50/ConvNeXt-T pair.

# O Corruption Robustness

We evaluate corruption robustness on CIFAR-100-C Hendrycks and Dietterich [2019], which applies 19 corruption types at 5 severity levels to the CIFAR-100 test set. We report mean corruption error (mCE), normalized so that same-architecture, same-normalization AdamW = 100%; lower is better.

Muon achieves the lowest mCE on two of four architecture configurations (ConvNeXt-T: 94.6%, ResNet-50-LN: 92.6%), with the robustness advantage most pronounced on layer-normalized architectures. On DeiT-S, the corrected Muon mCE (98.6%, using seed=2 baseline checkpoint with 62.26% clean accuracy) is marginally above SOAP (98.3%), indicating that the previously reported advantage was an artifact of using an enhanced checkpoint. On ResNet-50 with BatchNorm, differences among optimizers are marginal (mCE 98.6–100.5%), consistent with BatchNorm's known loss-landscape smoothing effect. Shampoo underperforms AdamW on DeiT-S (mCE 102.6%) and performs near-baseline on ResNet-50-BN (99.8%), but shows moderate improvement on ConvNeXt-T (98.1%) under multi-seed evaluation. Category-level analysis reveals that Muon's advantage concentrates in weather and digital corruptions (brightness, saturate, snow, frost, spatter), while noise corruptions (Gaussian, shot, impulse) show smaller inter-optimizer differences.

**Dose-response pattern in corruption robustness.** The GN(32) configuration extends the dose-response pattern observed in clean accuracy (Section 5.3) to corruption robustness. Under GN(32), all three spectral optimizers improve mean corruption accuracy by 5–7 pp relative to their BN counterparts (Muon: $+7.2$ pp, SOAP: $+5.1$ pp, Shampoo: $+5.0$ pp), while AdamW remains flat ($+0.2$ pp). SGD under GN(32) degrades to 14.36% mean corruption accuracy (mCE 112.9%), an 8.2 pp drop below GN(32) AdamW, mirroring the clean-accuracy collapse observed in the normalization-swap experiment. This three-way divergence (spectral methods improve, AdamW holds steady, SGD degrades) constitutes a dose-response pattern: as normalization smoothing is reduced from BN to GN(32), the optimizer's capacity for spectral preconditioning determines whether it can compensate for the lost implicit conditioning, in corruption robustness as in clean accuracy.

## O.1 Spectral Divergence Profile

We formalize the *stable rank divergence* $\rho$ as a post-hoc diagnostic for whether a spectral optimizer structurally modifies the weight spectrum.

Table 21: Corruption robustness on CIFAR-100-C. mCE (↓ is better) normalized so same-normalization AdamW = 100%. Retention = corruption accuracy / clean accuracy, measuring inherent robustness independent of clean performance level. mCE is computed over the standard 15 corruption types [Hendrycks and Dietterich, 2019].

| Architecture | Optimizer | Clean Acc (%) | Corr Acc (%) | mCE (%) | Retention (%) |
|---|---|---|---|---|---|
| ResNet-50 (BN) | SOAP | 77.28 | 22.74 | **98.6** | 29.4 |
| | Shampoo | 78.25 | 21.94 | 99.8 | 28.0 |
| | AdamW | 67.85 | 22.40 | 100.0 | 33.0 |
| | Muon | 76.88 | 21.59 | 100.5 | 28.1 |
| ResNet-50 (LN) | Muon | 78.52 | 27.44 | **92.6** | 34.9 |
| | AdamW | 50.51 | 22.93 | 100.0 | 45.4 |
| ConvNeXt-T | Muon | 74.59 | 26.66 | **94.6**$\pm$0.5 | 35.7 |
| | SOAP | 74.40 | 26.23 | 95.1$\pm$0.1 | 35.3 |
| | AdamW | 59.81 | 21.12 | 100.0 | 35.3 |
| | Shampoo | 70.57 | 23.55 | 98.1$\pm$0.4 | 33.4 |
| DeiT-S | Muon[‡] | 62.26 | 21.65 | 98.6 | 34.8 |
| | SOAP | 65.45 | 21.47 | **98.3** | 32.8 |
| | AdamW | 52.33 | 20.89 | 100.0 | 39.9 |
| | Shampoo | 63.53 | 18.77 | 102.6 | 29.5 |
| ResNet-50 (GN32) | Muon | 78.20 | 28.75 | **90.0** | 36.8 |
| | SOAP | 75.52 | 27.88 | 91.3 | 36.9 |
| | Shampoo | 74.79 | 26.96 | 92.6 | 36.0 |
| | AdamW | 55.73 | 22.58 | 100.0 | 40.5 |
| | SGD | 23.90 | 14.36 | 112.9 | 60.1 |

[‡] DeiT-S Muon: corrected using seed=2 baseline checkpoint (clean acc 62.26%); previous report (mCE 89.2%) used enhanced checkpoint.
ConvNeXt-T mCE: multi-seed means ($n = 3$ for Muon/SOAP, $n = 2$ for Shampoo). Shampoo's lower clean accuracy (70.4% vs 74.5%) may partially confound the mCE comparison through floor effects on noise corruptions.
Spectral preconditioning method: Muon and SOAP use Newton–Schulz iteration; Shampoo uses Kronecker factorization.

**Definition.** For a trained model with weight matrices $\{W_l\}_{l=1}^{L}$, the *stable rank* of layer $l$ is

$$\mathrm{SR}(W_l) \;=\; \frac{\|W_l\|_F^2}{\|W_l\|_2^2}\,, \tag{6}$$

where $\|\cdot\|_F$ and $\|\cdot\|_2$ denote the Frobenius and spectral norms. Given a spectral optimizer $\mathcal{S}$ and a baseline $\mathcal{B}$ (AdamW), the *stable rank divergence* is

$$\rho(\mathcal{S},\mathcal{B}) \;=\; \frac{1}{L}\sum_{l=1}^{L}\frac{\mathrm{SR}_{\mathcal{S}}(W_l)-\mathrm{SR}_{\mathcal{B}}(W_l)}{\mathrm{SR}_{\mathcal{B}}(W_l)}\,. \tag{7}$$

A positive $\rho$ indicates that the spectral optimizer distributes singular value mass more uniformly than AdamW (higher effective rank).

**Empirical findings.** We compute $\rho$ from best-checkpoint weights across three architecture configurations (Table 22). All spectral optimizers yield $\rho > 0$ ($n = 7/7$), confirming that they structurally increase the effective dimensionality of learned representations.

**Within-architecture ranking.** We observe a negative *observational* correlation between $\rho$ and $\Delta\mathrm{acc}$ within each architecture: on both ResNet-50 (BN) and ConvNeXt-T, the optimizer with the *lowest* $\rho$ achieves the *highest* accuracy gain (Kendall $\tau = -1.0$ on ConvNeXt-T). That is, the most effective spectral optimizers apply *minimal* spectral restructuring for maximal downstream improvement, rather than indiscriminately flattening the singular value spectrum.

**Normalization and spectral modification.** Comparing ResNet-50 under BatchNorm vs. LayerNorm reveals that Muon's $\rho$ increases from 2.10 (BN) to 6.94 (LN), a 3.3$\times$ increase. When BatchNorm's implicit spectral conditioning is removed, the optimizer must perform substantially

Table 22: Stable rank divergence $\rho$ and accuracy advantage $\Delta\mathrm{acc}$ over AdamW baseline. All checkpoints are best validation accuracy (seed 42).

| Architecture | Optimizer | $\rho$ | $\Delta$acc (%) |
|---|---|---|---|
| ResNet-50 (BN) | Muon | 2.10 | +8.89 |
| | SOAP | 3.92 | +9.34 |
| | Shampoo | 3.44 | +10.03 |
| ResNet-50 (LN) | Muon | 6.94 | +26.37 |
| ConvNeXt-T† | Muon | 4.53 | +11.78 |
| | SOAP | 7.79 | +8.46 |
| | Shampoo | 8.49 | +4.67 |

†Evaluated under enhanced training recipe.

more spectral restructuring to maintain performance. This is consistent with the dose-response pattern observed in Section 5.3: spectral optimizers compensate for the loss of normalization-induced regularization, and the magnitude of compensation scales with the degree of normalization removal.

**Limitations.** This analysis reports observational correlations from CIFAR-100 scale experiments (20 epoch budgets) and should not be interpreted causally. The ResNet-50 (LN) AdamW baseline (51.9%) reflects suboptimal hyperparameter transfer rather than inherent optimizer failure, inflating the apparent $\Delta\mathrm{acc}$. Validating $\rho$ as a *predictive* (rather than post-hoc) diagnostic requires computing stable ranks from early-training checkpoints and testing on larger-scale datasets, which we leave to future work.

### O.2 Tiny-ImageNet: SOAP and Shampoo Results

Table 23 reports Tiny-ImageNet accuracy for all three spectral optimizers across architectures, complementing the CIFAR-100 results in the main text.

Table 23: Tiny-ImageNet 20-epoch accuracy (%) for spectral optimizers. The ranking Muon $>$ SOAP $>$ Shampoo holds on ResNet-50 and ConvNeXt-T. On DeiT-S, all three spectral optimizers cluster within 3.5 pp, with SOAP slightly leading.

| Architecture | Muon | SOAP | Shampoo |
|---|---|---|---|
| ResNet-50 | 62.66 | 61.48 | 57.99 |
| DeiT-S | 48.02 | 51.49 | 50.58 |
| ConvNeXt-T | 58.92 | 57.46 | 49.98 |

The ranking Muon $>$ SOAP $>$ Shampoo holds on ResNet-50 and ConvNeXt-T. On DeiT-S, all three spectral optimizers cluster within 3.5 pp, with SOAP slightly leading, with the largest gap on ConvNeXt-T ($\Delta_{\text{Muon–Shampoo}} = 8.94$ pp). This cross-dataset replication strengthens the Newton-Schulz vs. Kronecker distinction reported in Section 5.

### O.3 Early Convergence Speed on ImageNet-1K

To assess scalability beyond CIFAR-100 and Tiny-ImageNet, we compare early training dynamics on ImageNet-1K (ResNet-50, single GPU).

Table 24: ImageNet-1K early convergence comparison: Muon reaches AdamW's accuracy in $\sim 4\times$ fewer epochs.

| Milestone | Muon epoch | AdamW epoch |
|---|---|---|
| $\sim$63.9% val acc | 5 | 20 |
| $\sim$67.1% val acc | 8 | 33 |

At epoch 5, Muon reaches 63.9% top-1 accuracy, matching AdamW's epoch-20 accuracy—a $4\times$ epoch speedup consistent with findings on smaller datasets (Table 18). At epoch 8, Muon achieves

67.1%, which AdamW requires ∼33 epochs to match. These results suggest the spectral advantage persists at ImageNet scale.

### O.4 Per-Layer Attention Analysis in DeiT-S

To determine whether Muon's spectral advantage concentrates in attention or MLP layers, we decompose the per-layer analysis of DeiT-S into four weight components per transformer block. Table 25 reports the stable rank ratio (Muon/AdamW) averaged across layers 1–11.

Table 25: Stable rank ratio (Muon/AdamW) by weight component in DeiT-S, averaged over layers 1–11.

| Component | Mean SR Ratio | Shape |
|---|---|---|
| MLP fc1 (fan-out) | 4.18 | $384 \to 1536$ |
| Q/K/V projection (square) | 3.49 | $384 \to 384$ each |
| Output projection (square) | 2.17 | $384 \to 384$ |
| MLP fc2 (fan-in) | 1.68 | $1536 \to 384$ |

Muon's advantage does *not* concentrate in attention layers: attention SR ratio ($2.83\times$) and MLP SR ratio ($2.93\times$) are comparable. The true driver is *matrix geometry*: the fc1 fan-out layer ($4.18\times$) shows the strongest spectral divergence, followed by square matrices (QKV: $3.49\times$, proj: $2.17\times$), with fan-in (fc2: $1.68\times$) lowest. The SR ratio is primarily driven by the fc1 fan-out layer, consistent with Muon's Newton-like update acting more aggressively when the matrix has more room for rank expansion along the output dimension.

AdamW concentrates 2–3$\times$ more energy in the top-5 singular values than Muon across all components (e.g., Q/K/V: Muon 4.65% vs. AdamW 11.97%), confirming that Muon produces more uniformly distributed singular value spectra regardless of functional role.

*Note: Based on a single seed; multi-seed validation would strengthen layer-level patterns.*

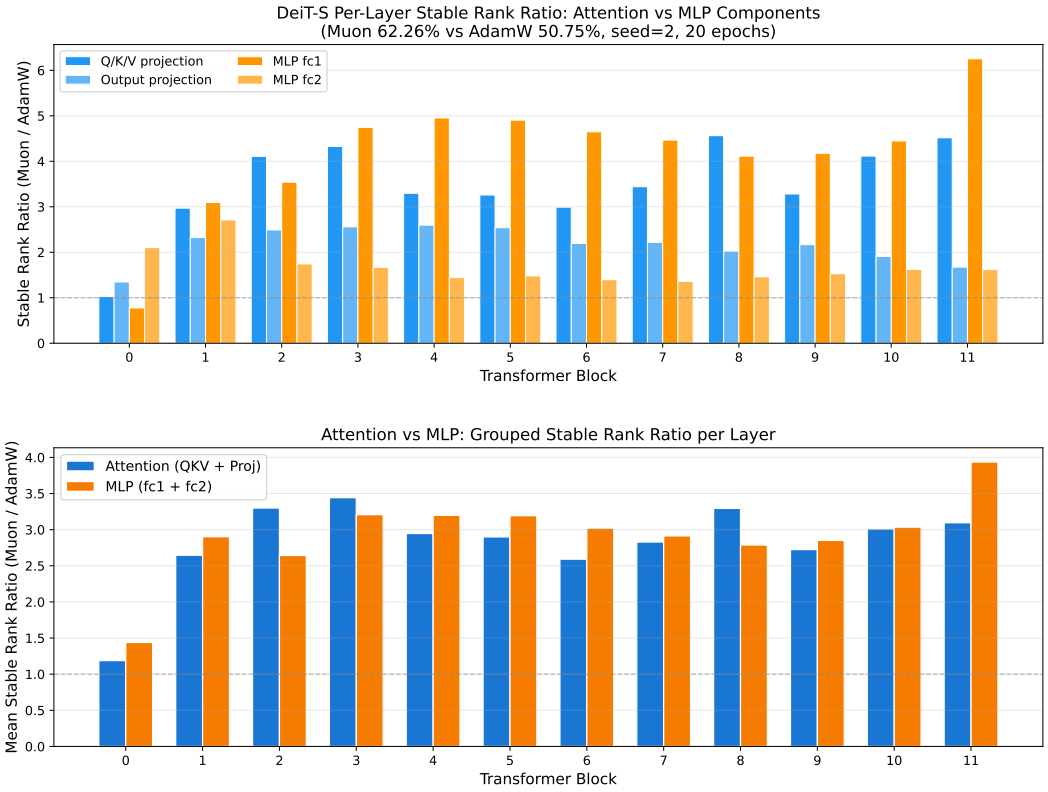



Figure 6: Per-layer stable rank ratio (Muon/AdamW) in DeiT-S, decomposed by component. The spectral divergence is primarily driven by the fc1 fan-out layer rather than functional role (attention vs. MLP).